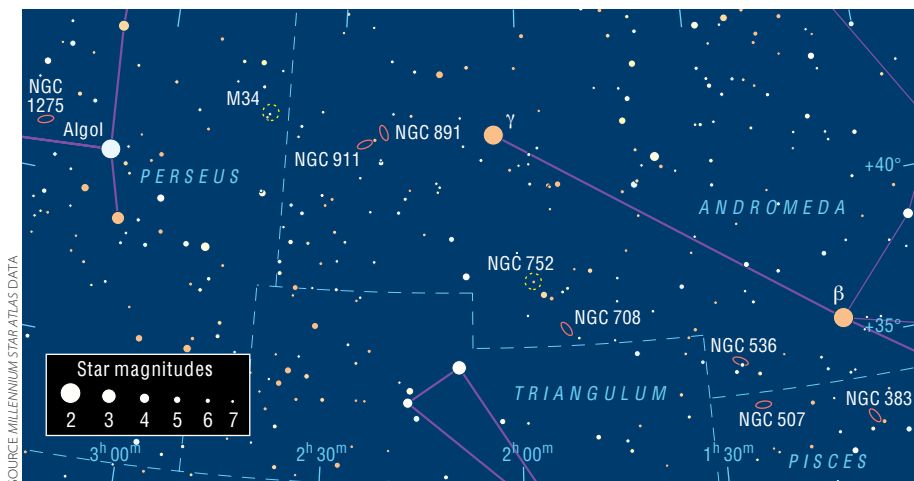




This key chart shows the locations of signpost galaxies in Steve Gottlieb's tour along the Pisces-Perseus Supercluster. The photographic finder charts that follow zoom in on small areas around these galaxies.

■ **deep-sky notebook** | *By Steve Gottlieb*

Exploring the Pisces-Perseus Supercluster

I enjoy seeing a lot of galaxy action in the eyepiece. From interacting pairs to rich clusters spilling over several eyepiece fields — the more galaxies in view, the merrier!

Galaxies like to clump together. Our own Milky Way and the Andromeda Galaxy are the leading members of a modest gathering known as the Local Group. A glance at a detailed star atlas reveals many other small groupings scattered across the sky, as well as a rich concentration of galaxies in the general direction of the giant ellipticals M84, M86, and M87 in Virgo. Here we find Edwin Hubble's Realm of the Nebulae, the Virgo Cluster, consisting of a few thousand galaxies centered about 50 to 60 million light-years away.

Big, bigger, biggest? How far does this hierarchy of clustering extend? In the late 1970s, large-scale redshift surveys revealed that clusters of galaxies are often linked together in chains to form enormous superclusters. Our Local Group resides in the outskirts of the Local Supercluster, first identified in the early 1950s by Gérard de Vaucouleurs, which has the Virgo Cluster as its core.

The winter sky is home to one of the largest known structures of all — the Pisces-Perseus Supercluster. This celestial filament stretches from the Perseus Galaxy

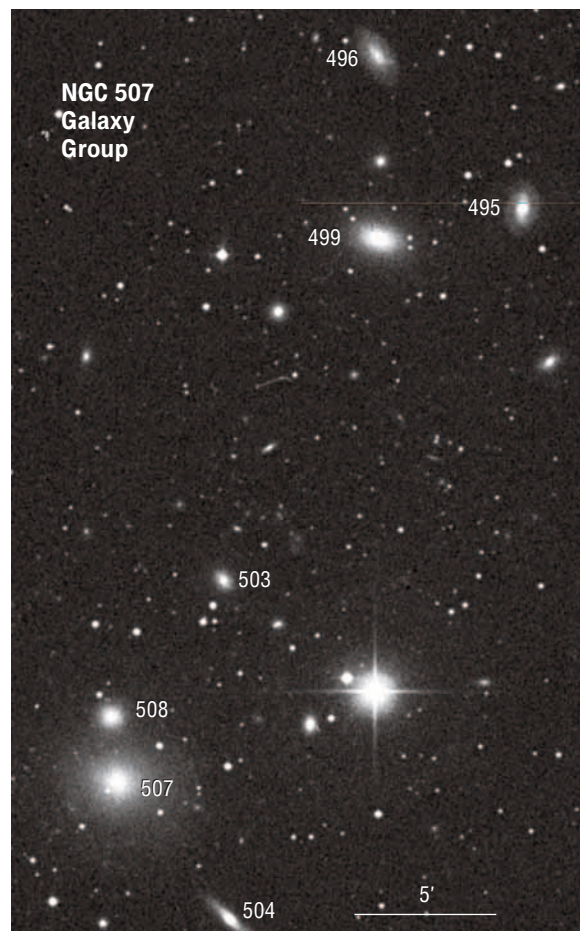
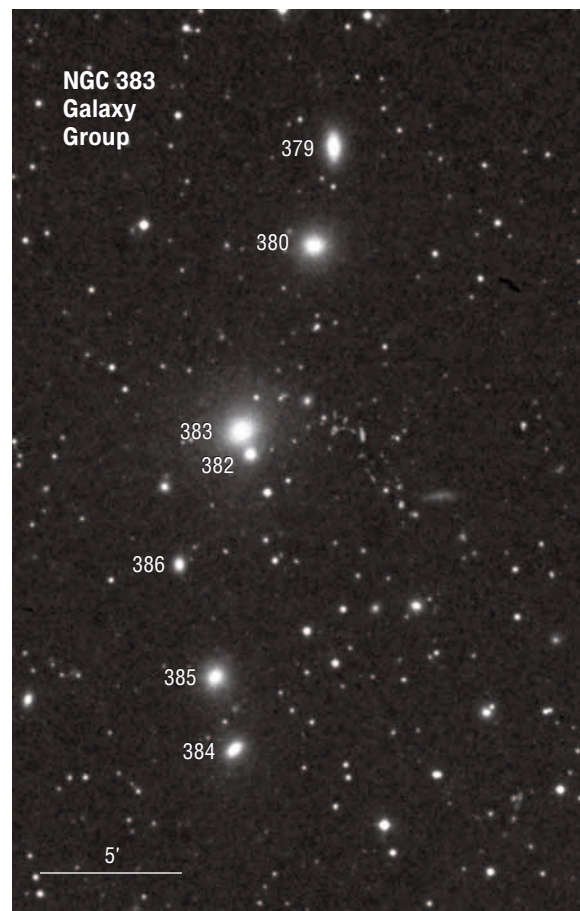
Cluster (Abell 426) through the rich clusters Abell 347 and Abell 262 in Andromeda on toward the NGC 507 and NGC 383 galaxy groups in northern Pisces.

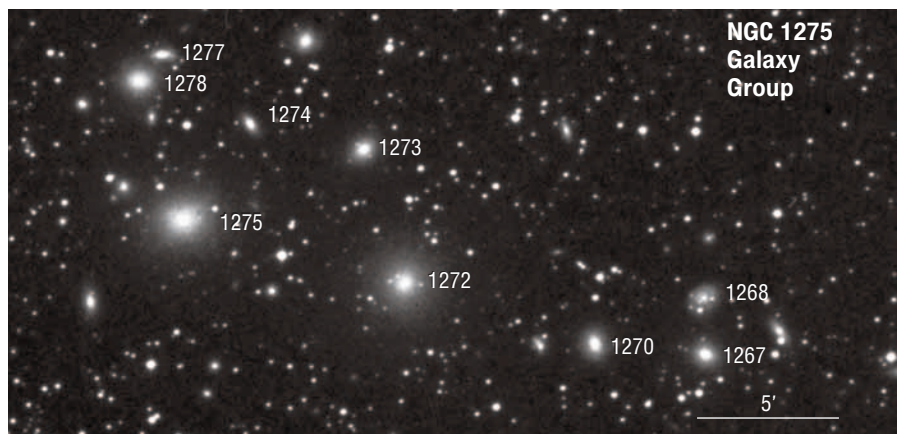
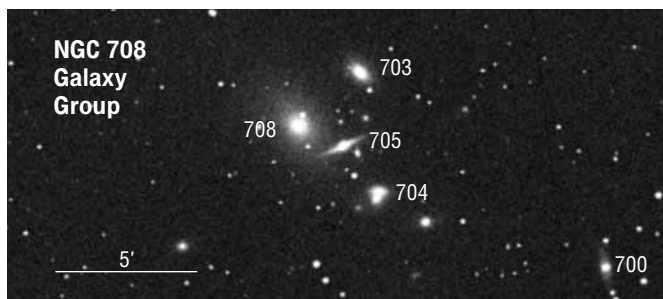
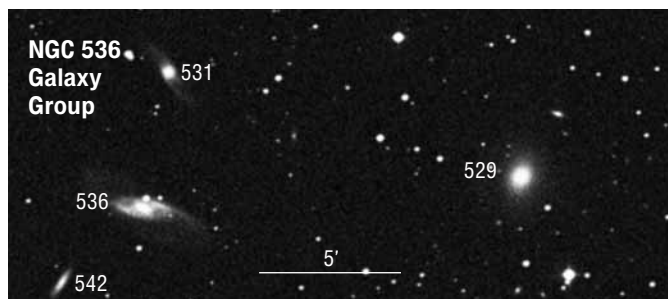
Even at a distance of 250 to 300 million light-years, this amazing chain of clusters spans more than 40° of sky. Can amateurs observe this structure? Fortunately, this thread from the large-scale fabric of the universe includes some of the brightest galaxy clusters.

Let's begin by picking up the trail near the southwest end, at the Pisces Group. The center of this cluster is dominated by 12.4-magnitude NGC 383, which appears in my 17.5-inch f/4.5 Dobsonian as a fairly bright, 1.5'-diameter oval with a large, bright core. See if you can spy its dim companion, NGC 382, dangling off the southwest side.

Extending north and south from this duo is a 14' chain of galaxies. At the north end, look for the pair NGC 379 and 380, separated by a mere 2'. Although similar in brightness and size, NGC 380 has a brighter core and a stellar nucleus, while NGC 379 has a smoother halo and is more elongated.

These finder charts and those that follow are based on Digitized Sky Survey images. North is up.





Marking the southern end of the chain is a slightly tighter pair, NGC 384 and 385. Both have round, 1'-diameter halos punctuated with bright cores.

Now meander about 45' northeast and you'll run across three trios that mark the path of the Pisces-Perseus ridge: NGC 392, 394, and 397; NGC 407, 410, and 414; and NGC 447, 449, and 451. Use a detailed atlas such as *Uranometria 2000.0* or, better yet, the *Millennium Star Atlas*.

Another 3° farther east, the NGC 507 cluster comes into view as we approach the Andromeda border. With my 17.5-inch scope, I've observed 15 galaxies within a 30' circle at the center of this group. One of the brightest members, NGC 499, has a strikingly bright core embedded in a 1.5' halo. Hovering nearby are several fainter galaxies, including NGC 495 and 496.

At 11th magnitude, NGC 507 dominates this clan with a 2' halo brightening sharply to an intense core. Just off its north edge is NGC 508, a small, faint elliptical. Only 4' southwest of NGC 507 is NGC 504, a small spindle three times as long as wide oriented southwest-northeast. Its axis is aligned with the core of NGC 507.

Continuing our trek northeast, we now cross the border into Andromeda to make a stop at the NGC 536 group, centered just 5' north-northeast of a 6.4-magnitude star. This quartet is cataloged in Paul Hickson's list of compact groups

as HCG 10 (*S&T*: March 1999, page 110). NGC 536 is a barred spiral with long, graceful arms. See if you can pick out the bright, elongated core and irregular halo. About 2' or 3' north and southeast are two fainter spirals, NGC 531 and 542. The group is completed by NGC 529, a bright lenticular galaxy 8' west.

A sparse trail of galaxies leads to our next highlight, Abell 262. This is a rich cluster tucked away in Andromeda 2° southwest of the big, loose open star cluster NGC 752. In the mid-1980s I collaborated with Canadian observers Paul Brown, Alister Ling, and James Lucyk to observe as many cluster members as possible. We ended up making 125 observations and identifying 65 cluster galaxies — not an easy task before computerized star charts and digitized sky surveys! Although most required a 12.5-inch or larger scope, more than a dozen were logged with an 8-inch Newtonian.

The core of Abell 262 includes a rich concentration of nine NGC galaxies, in-

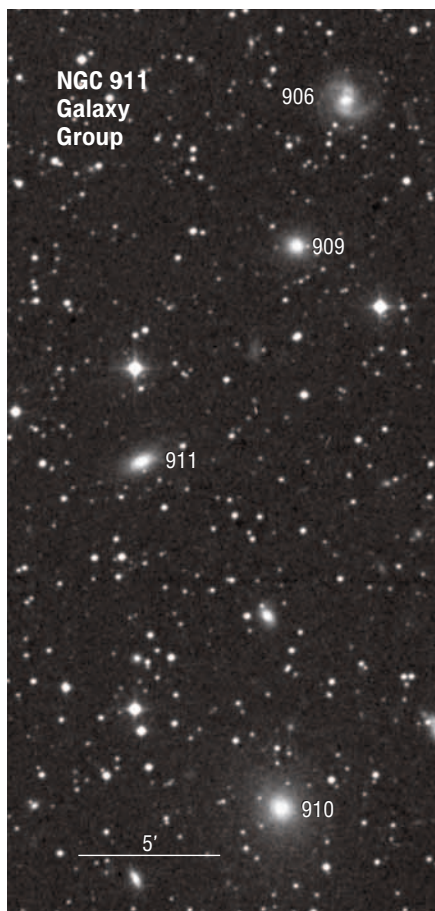
cluding the quartet of NGC 703, 704, 705, and 708, all crammed into a 3' circle. NGC 708, the largest of the lot, is visible as a fairly faint 1' glow with a broad, weak central concentration. A 15th-magnitude star appears attached at the northern edge of the galaxy's halo. Just 1' southwest is NGC 705, a small, faint lenticular with a 15th-magnitude star close by to the southwest. Another 1.3' southwest of NGC 705 is NGC 704, which is actually a double system with a starlike companion galaxy embedded in its southern edge. We finish up with NGC 703, a dim oval oriented southwest-northeast with another 15th-magnitude star just southwest.

Back along the supercluster ridge, we push on to our next destination, Abell 347. This cluster resides just 45' southeast of one of my favorites, the classic (and much nearer) edge-on galaxy NGC 891. The brightest members of Abell 347 are NGC 906, 910, and 911, run-of-the-mill 12th-to-13th-magnitude ovals aligned in a 20' string. Careful scrutiny of the area may yield as many as a dozen additional fainter members; a more detailed guide to Abell 347 appeared in last month's issue, page 125.

For the final leg of this journey, follow a trail of faint galaxies 10° due east past the bright open cluster M34 and brilliant Algol to the Perseus Galaxy Cluster, Abell 426. Its central region is a 40' string of galaxies running northeast from IC 310 to NGC 1281. A romp along this row is exhilarating. In my 17.5-inch I netted two dozen small, dim galaxies hiding in a tangled maze of Milky Way stars.

Key Galaxies of the Pisces-Perseus Supercluster

Name	R.A. (2000.0) Dec.	Mag. (V)	Atlas chart number		
			<i>Uranometria</i>	<i>Millennium</i>	Member of
NGC 383	1 ^h 07.4 ^m +32° 25'	12.4	91	147	Pisces Group
NGC 507	1 ^h 23.7 ^m +33° 15'	11.2	91	125	
NGC 536	1 ^h 26.4 ^m +34° 42'	12.4	91	125	HCG 10
NGC 708	1 ^h 52.8 ^m +36° 09'	12.7	92	123	Abell 262
NGC 911	2 ^h 25.7 ^m +41° 57'	12.7	62	100	Abell 347
NGC 1275	3 ^h 19.8 ^m +41° 31'	11.9	62	98	Abell 426



Abell 426 is ruled by its brightest member, the Seyfert galaxy NGC 1275, also known as Perseus A. I've glimpsed this peculiar 12th-magnitude galaxy with a 6-inch-aperture mask on my 17.5-inch. With full aperture, the galaxy's halo extends east to west 1.5' by 1.2' and contains a bright core with a starlike center.

Stretching west of NGC 1275 is a long chain of faint galaxies, including the cluster's second-brightest member, NGC 1272, with NGC 1270 and the close pair NGC 1267 and 1268 farther on. Take your time exploring this rich region. I've chased down nearly 60 members of this swarm.

Our long journey ends as we lose track of the supercluster ridge where it disappears behind obscuring dust clouds of the Milky Way. But it may not end here. Recent studies have traced the ridge through the Milky Way to where it appears to reemerge at Abell 779 in Lynx. If this is still the same string, it would be an impressive "Great Wall" of galaxies roughly 500 million light-years long!

Steve Gottlieb is a regular Deep-Sky Notebook contributor. His most recent article, "A Tour of Obscure Summer Globular Clusters," appeared in the August 2000 issue.



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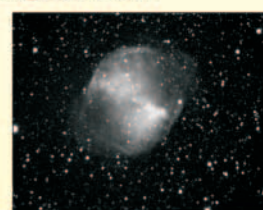


A unique patented invention by Starlight Xpress Ltd, 'S.T.A.R. 2000' will record full-frame CCD images while automatically guiding your telescope to sub-pixel accuracy. Unlike dual CCD autoguider systems, **S.T.A.R. 2000** uses the same CCD for both functions and allows you to guide on ANY object within the field being imaged! This eliminates the need to find a guide star with a tiny off-axis chip and makes it possible to guide on moving objects, such as comets. The **S.T.A.R. 2000** software takes advantage of the unique structure of the MX camera's 'interlaced' CCDs to download guider frames from one interlaced field, while integrating an image in the other field. This results in an output image without any lost area, at the expense of the need for a longer exposure. Of course, the extra exposure time is not a problem when the camera can guide itself!

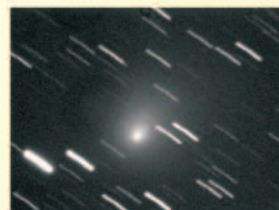


The NGC891& M27 images were 10 minute images using a C11 & an Mx916.

All images were taken using the patented S.T.A.R. 2000 Software.



The Comet Lee image was a 10 minute exposure with an Mx5.



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